

Case 2: Structural Vector AutoRegressive Models, Local Projections and Macro-economic Policy Analysis

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In this case you will explore structural vector autoregressive (VAR) models, local projections and impulse response analysis in R via a macro-economic application. You will use data on industrial production, the consumer price index and the federal funds rate to estimate a VAR on US data. You will learn (i) how to estimate VAR models, (ii) how to test for Granger causality, (iii) how to conduct impulse response analysis using VARs, (iv) what local projections (LPs) are and how to obtain impulse responses using LPs, finally (v) what structural VAR models and structural LPs are and how they can be used for policy analysis based on impulse response functions. A good introduction to VAR models (in addition to the rather mathematical introduction in Hamilton, 2020), is available in Kilian and Lütkepohl (2017), Chapter 2.

General Guidelines. The case study is deliberately formulated in a rather “open” way as you will have to “learn by doing” how to solve the tasks at hand. This document provides you with initial pointers for additional reading material (academic papers, software papers, ...) that may be relevant. You may resort to any other auxiliary material that helps you to solve this case. When presenting your case study in class, make sure to

- Prepare compact “lecture material” (for instance slides) when you need to introduce new time series concepts, models, estimators, ... to your fellow students.
- Prepare a careful analysis in R using readable scripts and intuitive figures, ... such that you can walk your fellow students through your software implementation. While you can make use of any auxiliary material, you should be able to explain all the code you generated in R (motivate use of certain packages and functions, know how the functions work, what arguments in the functions mean, ...)! You should also spend sufficient time on the interpretation of the output and the results. Make sure that your R scripts are in a format that allows for easy replication.
- Be transparent about, and take full responsibility for, any possible use of external tools, including generative AI. You may use such tools to support your work, but you must be able to explain, motivate and defend every (methodological) choice, every line of R code and every result live during the in-class presentations and Q&A. Answers that reveal a lack of genuine understanding of the underlying methodology, regardless of how they were produced, will be graded accordingly. Each team will also upload a reflection document on GenAI usage, see Canvas for more details.

Data. We will use monthly data from the Federal Reserve Bank of St Louis, publicly available at <https://research.stlouisfed.org/econ/mccracken/fred-databases/>. Start by collecting monthly data on industrial production, the consumer price index and the federal funds rate. Explain which variables from the FRED-MD database you selected to this end and motivate your choices! Consider ending your sample size at the end of 2019 to avoid influence of the pandemic. Alternatively: when you opt to include more recent years, make sure to carefully treat the pandemic years and discuss how your modeling choices affect your analysis. In this case study, you will estimate a trivariate VAR with these time series, in line with the VAR specification in Bernanke et al. (2005)¹

Data Exploration. Start by loading your data on the three variables into RStudio and by visually exploring the data. You should familiarize your fellow students with the data and its typical features. Consult the transformation codes provided by McCracken in the Appendix of McCracken and Ng (2016) to make your time series stationary. Do you transform your industrial production time series, and if so how? Do you transform your consumer price index time series, and if so how? Do you transform your federal funds rate time series, and if so how?

Unit Root Tests. To determine whether your time series are stationary or not you can combine economic reasoning with unit root tests. Perform appropriate unit root tests, using the Pantula principle, on each of the three series to determine their order of integration. Which unit root tests do you use? What do you conclude? Make a final choice on how to transform your time series such that you can use stationary time series as an input in your VAR model.

Estimating VAR Models and Selecting the Lag Order. Write down the model equation of the trivariate VAR you want to estimate. A first decision that needs to be made is how many lags you want to include in the VAR model. First, discuss popular approaches to determine the VAR order and summarize them to your fellow students. Second, find an appropriate R package to estimate VAR models and to select the order of the VAR based on the criterion/criteria you just discussed. You may want to consider Pfaff (2008) as a starting point. Present your analysis in R and carefully discuss how to interpret the results.

Once you selected the VAR order, then estimate the VAR model with the selected lag order. Write out the model equation of the estimated VAR and discuss your results. Finally, can you validate your VAR model? Which assumption(s) need to be satisfied? You likely have to run additional analyses/tests to this end. Motivate which ones and carefully discuss the results.

Granger Causality. A first key concept related to VAR models is Granger Causality. Start by re-explaining to your fellow students what it means when variable x Granger causes variable y . Should we interpret this as a true causal relations from x to y ?

¹We do not discuss the FAVAR specifications in the paper of Bernanke et al. (2005) in this case study or course. We will only focus on the trivariate VAR specification discussed as baseline model in this paper. Note that for interested students, it is good to know that you will further elaborate on this paper and the FAVAR model in the Big Data Econometrics course.

Second, find an appropriate R function in the package `vars` to perform the Granger Causality tests. Carefully write down the null and alternative hypothesis the function examines. What do you conclude?

Now assume you want to perform the six bivariate Granger causality tests in the trivariate VAR. Does the `vars` package allow you to perform these tests? If not, consider implementing the Granger causality tests yourself!

Impulse Response Functions for reduced-form VARs. A second key concept related to VAR models are Impulse Response Functions (IRF). Start by re-explaining to your fellow students what impulse response functions measure. Then, find an appropriate R function in the package `vars` to compute the IRFs. First focus on the point estimates of the IRFs. Carefully discuss the plots and make sure to explicitly indicate what the impulse and response is for each of the plots. In the Bernanke et al. (2005), the prize puzzle is mentioned in relation to one of the IRFs. Which one? Explain what this prize puzzle means. Does it occur in your analysis? Next, focus on the confidence interval around the IRFs. Carefully explain how the package computes the confidence interval and how this procedure works. Then comment on significance of your results.

Note that you likely transformed (some of) your variables to ensure they are stationary. Now, assume you want to obtain the IRFs for the original time series in levels. How can you obtain the IRFs for the time series in levels from the IRFs of the time series in differences? Have a look at Lütkepohl (2005), Chapter 2 Section 2.3.2 to this end. Carefully explain the procedure to obtain the IRFs for the original time series in levels to your fellow students. Then obtain the IRFs for the original time series in R. Discuss the plots carefully.

Finally, you have obtained IRFs for the standard, aka “reduced-form” specification of the VAR. Can we use the IRFs from the reduced-form VAR for structural analysis?

Reduced-form VARs versus Structural VARs. We will now turn to structural VARs (SVARs), see Kilian and Lütkepohl (2017) Chapter 4. Intuitively explain to your fellow students what the difference is between a reduced-form VAR and a structural VAR. Then provide the model equation of the SVAR and explain how it differs from the one of a reduced-form VAR. SVARs come together with a set of identification restrictions. What are these and what are commonly used economically-motivated identification restrictions, see Kilian and Lütkepohl (2017) Chapter 8.

Impulse Response Functions for recursive SVARs. Finally, we will consider recursively identified SVARs, by imposing restrictions on the ordering of the variables, and we will estimate corresponding IRFs, see Kilian and Lütkepohl (2017) Chapter 8. First, explain how the ordering of the variables leads to a recursively identified SVAR. Explain the impact of the ordering on the IRFs. Second, how would you order the three variables in the VAR? Motivate your choices.

Then, find an appropriate R package and function to estimate the SVAR and to obtain its corresponding IRFs. Carefully discuss the plots: how do they differ from the plots you have obtained earlier. How do you

interpret the results? How are the plots affected if you use a different ordering of the variables? Does it make sense to compare one ordering to another from a structural/causal perspective?

IRFs via VARs versus Local Projections. In the above, you have obtained the Impulse Response Functions by first estimation the VAR model. An alternative way to obtain IRFs is via the method of so called “Local Projections” (LP), see Jordà and Taylor (2025), Plagborg-Møller and Wolf (2021), Li et al. (2024). First, explain what LPs are and how we can obtain IRFs from them? How do the VAR-based approach and the LP-based approach compare, what are (dis)advantages of both? Second, find an appropriate implementation in R to obtain reduced-form IRFs via local projections. Compare your reduced-form VAR-based results to the LP-based results and discuss.

Structural IRFs using Local Projections. Structural IRFs can also be obtained using LPs, see Jordà and Taylor (2025). What type of identification strategies can be used in the LP framework? Choose one of these identification strategies and find an appropriate implementation in R. Compare your reduced-form VAR-based results to the LP-based results and discuss.

Using Shock Proxies. In the above, you have identified the structural shocks by imposing a recursive ordering. Recursive identification is the basis for more advanced identification schemes, the most commonly used of which in modern macroeconometrics are so called “shock proxies”, see Stock and Watson (2018). The US Monetary Policy Event Study Database (USMPD, <https://www.frbsf.org/research-and-insights/data-and-indicators/us-monetary-policy-event-study-database/>) is one example of such proxies. First, explain what a shock proxy is and how it identifies the structural shock. Which conditions does the proxy have to satisfy for this to work, and what can go wrong if they fail? Second, find an appropriate implementation in R, either among the packages you have used before or among more specialised ones, and obtain IRFs by using the proxy either as an internal instrument in the SVAR (Plagborg-Møller and Wolf, 2021) or in a local projection via LP-IV (Stock and Watson, 2018). Compare your proxy-based results to the ones you obtained before and discuss. Which estimates do you consider more trustworthy, and why?

Final Recommendations. Conclude with your lessons learned, tips & tricks and final recommendations for this case.

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